

RYAZHSK, Russian Federation

WMO: 278350

Lat: 53.717N

Lon: 40.117E

Elev: 417

StdP: 14.48

Time Zone: 3.00 (E03)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.9	-9.3	-20.1	1.9	-14.7	-14.2	2.6	-8.8	17.0	21.9	14.1	27.9	1.3	20	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	90.6	67.8	86.5	67.3	82.9	66.1	70.6	84.1	69.3	82.7	67.8	80.3	4.7	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.4	98.6	75.9	64.9	93.5	74.2	63.4	88.9	72.8	34.7	83.7	33.6	82.9	32.3	80.5	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	11.5	10.0	-23.5	93.2	6.8	4.7	-28.4	96.5	-32.3	99.3	-36.1	101.9	-41.0	105.3
			-23.8	73.9	6.7	2.4	-28.6	75.7	-32.4	77.1	-36.2	78.4	-41.0	80.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	15.6	17.5	27.2	43.7	59.4	64.0	68.9	66.6	54.9	43.4	32.0	24.4	(6)
(7)		DBStd	20.83	13.70	13.01	9.61	8.49	8.05	6.77	6.58	7.69	6.49	7.79	8.53	10.49	(7)
(8)		HDD50	4521	1068	911	707	230	15	1	0	23	231	540	795	(8)	
(9)		HDD65	8319	1533	1331	1172	640	209	100	35	72	310	668	989	1260	(9)
(10)		CDD50	2064	0	0	0	40	307	420	585	514	170	27	1	0	(10)
(11)		CDD65	389	0	0	0	0	36	69	155	122	7	0	0	0	(11)
(12)		CDH74	4256	0	0	0	15	476	676	1642	1371	76	0	0	0	(12)
(13)		CDH80	1537	0	0	0	2	114	194	632	584	11	0	0	0	(13)
(14)	Wind	WSAvg	4.2	4.5	4.6	4.8	4.5	4.2	4.1	3.2	3.5	3.4	4.1	4.6	4.9	(14)
(15)	Precipitation	PrecAvg	21.4	1.4	1.1	1.0	1.3	1.8	2.2	3.1	2.1	1.9	1.9	1.8	1.7	(15)
(16)		PrecMax	29.2	4.1	3.1	2.1	3.5	5.1	8.3	6.3	5.3	6.3	5.8	3.1	3.3	(16)
(17)		PrecMin	15.2	0.2	0.0	0.2	0.2	0.0	0.1	0.4	0.1	0.1	0.1	0.7	0.7	(17)
(18)		PrecStd	3.8	0.8	0.7	0.5	0.7	1.3	1.6	1.6	1.1	1.3	1.2	0.7	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	37.8	56.7	75.8	87.4	89.5	98.3	99.6	81.4	68.1	54.2	45.3	(19)
(20)		MCWB	36.3	35.4	48.8	55.0	66.3	65.6	67.8	68.5	63.3	56.2	50.3	43.5	(20)	
(21)		2%	DB	35.9	36.0	47.4	68.7	83.2	85.2	90.6	91.3	75.3	62.6	49.9	41.9	(21)
(22)		MCWB	34.5	34.1	40.4	53.0	63.6	66.4	68.2	67.8	60.3	54.9	47.0	40.8	(22)	
(23)		5%	DB	34.2	34.7	41.9	64.6	79.4	81.5	86.6	86.3	70.8	57.9	46.6	39.1	(23)
(24)		MCWB	32.9	33.2	36.7	51.6	61.9	66.1	68.0	67.6	58.8	52.8	44.4	38.0	(24)	
(25)		10%	DB	32.8	33.3	38.6	59.9	75.5	77.4	82.6	81.1	66.9	54.5	43.2	35.8	(25)
(26)		MCWB	31.8	32.2	35.5	48.7	60.7	64.1	66.9	66.0	56.9	50.5	41.2	34.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.0	36.4	47.4	57.8	68.8	71.2	72.7	71.6	64.8	58.7	51.0	43.9	(27)
(28)		MCDWB	37.5	37.2	56.0	69.5	82.3	82.4	85.7	85.7	75.2	64.1	53.5	44.8	(28)	
(29)		2%	WB	34.7	34.7	41.4	54.7	65.7	69.2	71.0	70.1	62.1	56.0	47.5	40.8	(29)
(30)		MCDWB	35.5	35.5	46.1	65.3	79.6	81.0	84.2	85.8	72.8	60.9	49.4	41.6	(30)	
(31)		5%	WB	33.2	33.4	37.7	52.4	63.5	67.3	69.7	68.4	60.2	53.2	44.6	38.0	(31)
(32)		MCDWB	34.0	34.3	40.8	62.2	76.1	77.5	82.5	83.9	68.4	57.2	46.5	39.0	(32)	
(33)		10%	WB	31.8	32.2	35.8	49.8	61.4	65.5	68.2	67.0	58.3	50.8	41.3	34.9	(33)
(34)		MCDWB	32.8	33.4	38.0	58.4	72.9	74.8	80.2	80.1	64.6	54.2	43.0	35.6	(34)	
(35)	Mean Daily Temperature Range	MDBR	10.9	12.3	14.3	18.6	21.4	19.1	20.7	21.3	17.9	13.2	9.0	8.4	(35)	
(36)		5% DB	MCDBR	6.7	9.0	17.7	26.6	26.8	24.5	26.5	27.8	25.0	18.2	13.7	6.8	(36)
(37)		MCWBR	6.8	8.1	11.4	13.4	11.6	10.3	9.1	10.1	12.4	11.4	11.2	6.7	(37)	
(38)		5% WB	MCDBR	6.4	7.9	15.1	23.1	24.1	21.5	22.7	24.6	21.1	16.0	11.6	6.5	(38)
(39)		MCWBR	6.5	7.5	10.4	12.7	11.8	10.3	9.3	10.3	11.0	11.1	10.3	6.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.296	0.336	0.357	0.428	0.393	0.388	0.419	0.439	0.385	0.361	0.331	0.299	(40)	
(41)		taud	2.109	2.024	2.068	2.089	2.306	2.343	2.272	2.201	2.332	2.324	2.229	2.124	(41)	
(42)		Ebn at Noon	201	229	255	252	271	273	261	247	247	225	189	174	(42)	
(43)		Edh at Noon	22	34	40	45	38	37	39	40	31	25	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	263	574	1056	1330	1793	1835	1821	1504	978	480	205	143	(44)	
(45)		RadStd	53	68	138	112	151	194	161	147	127	70	42	44	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (7 neighbors)	+1.40	N/A	N/A	N/A	N/A	N/A	-351	+230	(47)

Nomenclature: See separate page